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Adoption of Sobol's analysis method improved the application of a coupled primary and secondary infection grape downy mildew model in northern China

Wang Hui [a], Zhang Wei [a], Simone Bregaglio [b], Yu Shuyi [c], Li Xinghong [a], Li Yonghua [a], Liang Chunhao [c,*], Yan Jiye [a,*]

[a] Beijing Key Laboratory of Environment Friendly Management on Fruit Diseases and Pests in North China, Institute of Plant Protection, Beijing Academy of Agriculture and Forestry Sciences, Beijing, China
[b] CREA - Council for Agricultural Research and Economics, Research Centre for Agriculture and Environment, Bologna, Italy
[c] Institute of Plant Protection, Liaoning Academy of Agriculture Sciences, Liaoning, China

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ABSTRACT

Grape downy mildew is a major biotic constraint to grapevine production worldwide, and its impact is influenced by environmental conditions and varietal susceptibility. Here, we proposed a grape downy mildew model to optimize plant disease management and its application on northern Chinese grapevine areas. A primary and a secondary infection model of *P. viticola* were integrated to estimate the date of symptoms on set and disease incidence, to give the firSt brEakout and incidEnce of grape Downy Mildew model (SEE_DM). The experimental data for model calibration were collected on two grapevine varietals with high and moderate susceptibility to downy mildew grown in a multiyear (2009, 2011–2019) and multisite (Beijing, Shenyang and Yantai) trial. A model sensitivity analysis (Sobol analysis) drove the selection of the subset of relevant parameters to be adjusted in calibration. The model adequately reproduced the observed downy mildew incidence, obtaining high R^2 (0.89), Nash-Sutcliffe modelling efficiency (0.72), and low RMSE (9–16%). The model correctly estimated the disease onset date in all conditions but two (late by 5 and 13 days), demonstrating to be a valid tool to take timely decision to limit the time course of the disease. The model was then projected in three sites under varying temperature and rainfall to analyses their effects on the trends of grape downy mildew incidence, using a variety with moderate susceptibility. This study highlights that Northern China is gradually becoming more suitable for grape downy mildew infections, along with warmer temperatures and more frequent rainfall.

1. Introduction

Grape downy mildew (*Plasmopara viticola*) is a widely studied disease, as it is one of the most damaging grapevine productions worldwide. The pathogen originated in the northeastern grapevine area in the USA (Carisse, 2016; Gessler, 2011) and was introduced in France in 1878; from where it spread widely throughout Europe (Ash, 2000). The earliest relevant literature published in China involved a description of the symptomatology of grape downy mildew in 1982 (Wang, 1982). Its foliar symptomatology is constituted by yellow, oily spots with irregular shapes in the early onset period; these spots are caused by oospores overwintering in the soil and on infected leaves. White sporangia are then produced in the bottom leaf surface when temperature and relative humidity conditions are conducive, causing secondary infections on uninfected tissues through wind and splashing rain. This polycyclic disease is threatening grapevine production in 420,800 ha in northern China (China Statistical Institute, 2022); in particular, the frequently variable climate conditions identified in recent years have contributed to a further spread and heavier disease incidence (Chaloner et al., 2021).

Reliable disease-warning models are essential for understanding the response of grape downy mildew pathogen to weather variability to aid decision-making to control the disease. Although many empirical models have been developed and documented to provide support to counteract grape downy mildew, most of these approaches were

* Corresponding authors at: Beijing Key Laboratory of Environment Friendly Management on Fruit Diseases and Pests in North China, Institute of Plant Protection, Beijing Academy of Agriculture and Forestry Sciences, Beijing, China (Yan Jiye).
E-mail addresses: liangchunhao99@126.com (L. Chunhao), jiyeyan@vip.163.com (Y. Jiye).

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developed using past observations and empirical relationships (e.g., MILVIT, Muckensturm et al., 1990; POM, Tran Manh Sung, 1990). Process-based modeling techniques can be used to reproduce the nonlinear processes underlying the biological cycles of pathogens in response to variable environmental conditions (Wang et al., 2021b). These models describe the conditions under which *P. viticola* can mature or germinate; this information can be merged into models to reproduce the dynamic pathogen infection process (Rossi et al., 2009; Park et al., 1997). Among the existing models, the Rossi model has been promoted and validated over large areas and has become a mature tool for forecasting primary grape downy mildew infections (Rossi et al., 2009).

The Rossi model has been used to simulate primary inoculum seasons in southern China (Yi et al., 2019), assuming the extension of the primary inoculum season until early May in Hunan Province. The assessment of Rossi model in northern China is scarce, several special steps need to be optimized to adopt to local condition, including oospore maturation, germination, sporangia survival, zoospores release and dispersal and inoculation. Moreover, further evaluations of the capability to predict the disease incidence dynamics with simulation models are lacking. To this end, we i) optimized the Rossi model with by adjusting parameter values related to climate conditions and varietal characteristics and ii) we coupled it with a model of secondary infection stage of *P. viticola*, which considered the weather suitability for the sporulation, and the efficiency of spore's release and dispersal to simulate grape downy mildew incidence. The new model estimating the date of symptom onset and predicting the time course of grape downy mildew in northern China was termed SEE_DM (firSt brEakout and incidEnce of grape Downy Mildew).

SEE_DM was calibrated with field data collected in multiple years (2009, 2011–2019) and sites (Beijing, Yantai and Shenyang), providing quantitative figurers to support fungicide scheduling and decision-making processes of Chinese grapevine growers.

2. Materials and methods

2.1. Research overview

The workflow of this modeling study is presented in Fig. 1. Two data sources were used as model inputs: 1) field data collected in 2011, 2013 and 2015 in Beijing (Cabernet Sauvignon), in 2012–2019 in Shenyang (Centennial Seedless), and in 2009 in Yantai (Cabernet Sauvignon); and 2) historical weather data (2000 – 2020), provided by the Global Land Data Assimilation System (GLDAS) Noah Land Surface Model, and derived from the National Aeronautics and Space Administration (NASA) data portal (https://disc.gsfc.nasa.gov).

The SEE_DM structure presents two modules, the primary and secondary infection modules, which simulated the epidemiological processes of oospore maturation, germination, sporangia survival, zoospore release and dispersal (primary infection module) and infection, inoculation, sporulation, zoospore release and dispersal (secondary infection module). The influence of model parameters in modulating model outputs was ranked via Sobol's method to identify the most relevant (Vazquez-Cruz, et al., 2014). These parameters were automatically optimized to minimize the errors between observed and simulated disease onset date and incidence. The calibrated model was able to fit variable dynamics of downy mildew incidence in multisite and multivarietal trials. A climatic scenario analysis has been performed including changes in annual rainfall and average annual temperature to assess the expected changes in trends of grape downy mildew incidence.

2.2. Modeling grape downy mildew

The SEE_DM indicator was realized within the framework of the Rossi model (Rossi et al., 2008) to represent the primary infection process, thus allowing the modeling of the secondary infection process to be extended and the local conditions in northern China to be adopted by the

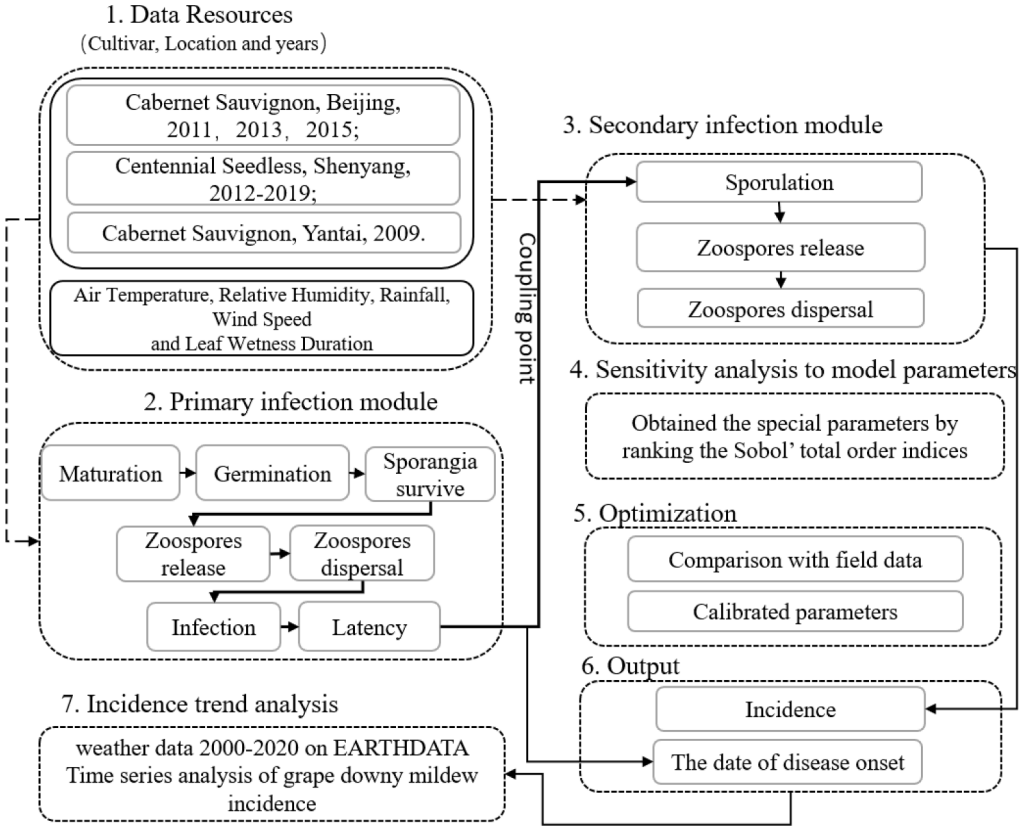


Fig. 1. The design, optimization and analysis of the grape downy mildew model used in this study.

algorithms implemented in the model components.

The primary infection is executed from Jan 1st, assumed as the moment when oospores break from their dormant state exhibited throughout the previous winter. Oospore maturation requires a hydro-thermal accumulation period, which can be calculated according to an equation (*f* (T_h and M_h), unitless, Eqs. (2)–(3) in Rossi, et al., 2008) expressing the hourly average air temperature (T_h) and the moisture state of the leaf-litter-holding oospores (M_h = 1 or 0). Oospore maturation starts and finishes when *f* (T_h, M_h) is equal to 1.3 and 8.6, respectively. Oospore germination is triggered when the rainfall amount is equal to or over 2 mm and is terminated within the first dry hour following rainfall. The sporangia survival state is evaluated depending on the T_h and relative humidity (RH) conditions, as shown in Eq. (6) from Rossi et al., 2008. Zoospores are released when the wetness duration (WD) passes the required duration (*f*(TWD), which is unitless as shown in Eq. (7) in Rossi, et al., 2008). Hourly rainfall above or equal to 0.2 mm facilitates zoospore dispersal. Zoospores have the opportunity to infect leaves under favorable conditions, i.e., when the leaf wetness duration (WD_h) multiplied by the average air temperature under the duration (TWD_h) is equal or over 60 (°C*hour). After leaves are infected, oil spots appear following the inoculation period, corresponding to the first grape downy mildew breakout. The inoculation period then ends along with the time interval (*f1*(T_h) and *f2*(T_h), both of which are unitless, as shown in Eqs. (8)–(9) in Rossi, et al., 2008).

Moreover, the grape growth stage determines the probability of the primary infection occurring (Park, 1997). Based on observations conducted over 12 years at three sites, the flowering stage determines the occurrence of grape downy mildew infections in northern China. The phenological stage (GP, Eqs. (1)–(2)) was obtained herein using a piecewise equation comprising the daily average temperature (T_{day}) and base temperature (T_{base} = 10 °C, Nogueira Júnior et al., 2018). The average GP values simulated over multiple years at each site were then used as the threshold range during the flowering stage (GP_r, see Table 1 in Section 2.4).

$$DD(T_{day}) = \begin{cases} T_{day} - T_{base} if T_{day} \geq T_{base} \\ 0 else \end{cases} \quad (1)$$

$$GP = \sum_1^n DD(T_{day}) \geq GP_r \quad (2)$$

After the end of the inoculation period, secondary infection process started running. Sporangia are produced on the undersides of the leaf; this period is called the sporulation stage. The sporulation efficiency (SP_h, unitless, ranging from 0 to 1, Eq. (3)) was computed as a function of the T_h and RH. T_{max_sp}, T_{min_sp}, and T_{opt_sp} represent the maximum, minimum and optimal air temperatures for sporulation, respectively, and RH_{min} is the minimum RH demand.

$$SP_h = \begin{cases} \left(\frac{T_{maxSP} - T_h}{T_{maxSP} - T_{optSP}}\right)\left(\frac{T_h - T_{minSP}}{T_{optSP} - T_{minSP}}\right)^{\left(T_{optSP} - T_{minSP}\right)/T_{maxSP} - T_{optSP}} & if RH > RH_{min} \\ 0 else \end{cases} \quad (3)$$

Then, zoospores are released when the wetness duration threshold is met (*f*(TWD), unitless, Eq. (7) in Rossi, et al., 2008). Zoospore dispersal occurs when the rainfall threshold ($Rain_{th}$ = 0.2 mm) is reached. The extent of splashing rain (*Rain*, mm) determines the dispersal probability (Wang et al., 2021b), and this factor was computed herein using Eq. (4), which considers the leaf area index (LAI, $m^2\ m^{-2}$) and the precipitation required to detach 50% of the spores at the given LAI ($Rain_{50}$, mm). A degree-day (DD) equation with a daily average temperature condition boundary ($T_{ave} \geq 10$°C) referenced from the Williams and Ayars (2005) study was used in this work to calculate the LAI (Eq. (5)).

$$Raineff = \frac{Rain}{Rain_{50} \times LAI + Rain} \quad (4)$$

$$LAI = -2.6 + 6.63*(1 - exp^{-0.0042*DD})T_{ave} \geq 10^{\circ}C \quad (5)$$

The incidence of grape downy mildew (I_h in Eq. (6)) was calculated using an equation containing the maximum hourly increase incidence (MI_h, unitless, 0–1), the sporulation efficiency (SP, unitless, 0–1), the dispersal probability caused by splashing rain (Raineff, unitless, 0–1) and an empirical coefficient used to simulate the susceptibility levels of various grape varietals to grape downy mildew (Sus, unitless, 0–1).

$$I_h = \begin{cases} \sum_{i=1}^{n=24} (MI_h \times SP \times (Raineff + Sus)) & if releaseevent = true and sprayevent = false \\ 0 elsewhere \end{cases} \quad (6)$$

Furthermore, fungicide spraying delays or impedes the increased incidence of grape downy mildew, and this condition is an unignored condition for when studying disease epidemiology. Spraying events (true = 1 or false = 0) were considered in the SEE_DM model based on the pesticide schedule applied in Beijing. We assumed that the increased incidence (I_h) of grape downy mildew would remain at zero for 7 days following a spray event, based on the schedule in which fungicide is applied every seven days (Eq. (7)).

$$\sum_{i=1}^{24*7*i} I_h = \begin{cases} \sum_{i=1}^{24*7*i} I_h if Sparyevent = 0 \\ 0 else \end{cases} \quad (7)$$

Moreover, the susceptibility coefficient (Sus in Eq. (6)) was set to 1 for Cabernet Sauvignon, a highly susceptible variety, as determined from a previous ranking of the susceptibility levels of sixteen wine grapes and nineteen table grapes grown in China (Sha et al., 2007). For the Centennial Seedless variety, the same coefficient was optimized

Table 1
Number of years and varietals used in the model evaluation process as well as the disease onset dates and climate conditions at the study sites.

Year	Site	Varietal	Disease onset	AT[c]/ ℃	RH/ %	AR[d]/ mm	Spray	Maturation
2009	Yantai	Cabernet S. [a]	May 26th	13.35	60.31	529.2	no	September 3th
2011, 2013, 2015	Beijing	Cabernet S.	July 1st −25th	12.68	72.01	586.6	yes	September 4th-17th
2012–2019	Shen-yang	Centennial S. [b]	June 25th-July 17th	8.79	61.17	743.54	no	August 24th-30th

[a] Cabernet S.: Cabernet Sauvignon; [b] Centennial S.: Centennial Seedless; [c] average annual temperature; and [d] annual rainfall.

within the [0,1] range.

2.3. Field experiments

The weather inputs required by the SEE_DM model were recorded by local stations located at the Bo Longbao winery in Beijing (116.02°E, 39.75°N), the Liaoning Academy of Agriculture Sciences in Shenyang (123.56°E, 41.83°N) and the Zhangyu winery in Yantai (121.18°E, 37.58°N). The average temperature, relative humidity and annual rainfall data are listed in Table 1.

The variety cultivates in Beijing is Cabernet Sauvignon; this varietal is grown with a row spacing of 1 m × 3 m. Chemical controls were conducted following field managers' standards; the grapes were sprayed twice monthly (Table S1 in the supplementary materials). In Shenyang and Yantai, Centennial Seedless (0.6 m × 0.5 m) and Cabernet Sauvignon (1 m × 1.2 m) varietals were planted, respectively, and no chemical control against grape downy mildew was applied during the growing season.

The investigations began when 5–6 grape leaves had unfolded (ELS = 12), and 200 plants were randomly observed in each plot every 2–3 days in 2 plots (80 m × 3 m) in the Beijing and Yantai experimental fields and 3 plots (15 m × 5 m) in Shen Yang. After oil spots were observed on the grape plants, 15 plants were investigated weekly at five sampling points in Beijing and Yantai and every 2–7 days in Shenyang until maturation (Table 1). The incidence of grape downy mildew (the proportion of infected leaves among all leaves on a one-year-old branch) and the pesticide schedule were both recorded.

2.4. Sensitivity analysis to model parameters

Performing a sensitivity analysis on model parameters is essential for obtaining the special parameters that help adapt the model responses to pathogen/varietal/weather differences. The initial values of all parameters were set to be consistent for the three analyzed sites and two analyzed varietals. The parameters of primary infection model were taken from the Rossi et al., 2008 study. In the secondary infection model, initial values of parameters were varied between plausible ranges (Table 2).

The relevance of each parameter was ranked using Sobol's total order index based on the modelled output variance considering all interactions between each one parameter and the others (SA, Sobol, 1993). The parameters with low total order index values were fixed, whereas the most relevant were optimized (Nossent et al., 2011).

To identify key parameters, the SA repeatability test was conducted with sample number×(2 + parameter number) iterations by increasing the sample number of each parameter from 4 to 40 until the total order index exhibited a weak fluctuation. The bootstrapping sample technique was used to resample the boundary conditions of each parameter (shown in Table 2) to reach the required sample number.

2.5. Model calibration and evaluation

The automatic model calibration was performed via the gradient descent method (Metzler, 2007), using the mean square difference between observations and simulations as objective function. The parameters were optimized between the boundary conditions obtained from literature sources (Table 2). Two parameters were set to default values due to limitations in available information (i.e., maximum incidence increases hourly and flowering phase threshold).

The calibration dataset was randomly sampled from the 12 experimental trials and comprised experiments in Beijing in 2011 and 2015 and in Shenyang in 2012, 2013, 2015 and 2017. The remaining datasets (Beijing in 2013, in Yantai in 2009, and in Shenyang in 2014, 2016, 2018 and 2019) were used to evaluate model performances. Each dataset included information on two grapevine varieties.

The two SEE_DM outputs were compared with the observations. The evaluation metrics were the difference in observed and predicted symptoms onset date (days), and the mean bias error (MBE, min. = -∞; opt. = 0; max. = +∞), root mean square error (RMSE, min. and opt. = 0; max. = +∞), determination coefficient (R^2, max. and opt. = 1; min. = 0) and Nash-Sutcliffe modeling efficiency (EF, min. = -∞, opt. and max. = 1) to evaluate model performances concerning disease incidence.

Table 2
Model parameter values, units, and literature sources.

Process	Parameter	Range	Unit	Value	Source [a]
Primary infection	Rain for Maturation	[0–5]	mm	2.30	1,2
	VPD for Maturation	[4–5.2]	hPa	5.20	1
	Hydro-thermal Accumulation for Maturation/ HT	[1.3–8.6]	unitless	6.8	3
	Temperature for Germination	[0–13]	℃	5.00	4–7
	Rain for Germination	[0–10]	mm	2.50	4,7
	Rain for Dispersal	[0.2–2]	mm	1.90	8
	$WD_h \times TWD_h$/ Threshold of Infection	[45–88]	℃xhour	83.00	9–12
	flowering phase threshold		unitless	[90–400]	Default
	Base Temp for Grape Growth	[10–11]	unitless	10.00	18–19
Secondary infection	Optimum Temperature for Sporulation/ T_{opt_sp}	[15–29]	℃	15.00	9,22
	Minimum Temperature for Sporulation /T_{min_sp}	[5–13]	℃	5.50	11,13
	Maximum Temperature for Sporulation /T_{max_sp}	[29–35]	℃	34.00	13–14
	Minimum Relative Humidity for Sporulation /RH_{min}	[60–98]	%	85.0	10,15
	Rain for 50% Detachment/ $Rain_{50}$	[0.2–3]	mm	0.20	16–17
	The Maximum Incidence Increase Hourly/MI_h		%	[0.002–0.005]	Default
	Grape varietal susceptive level to downy mildew	[0–1]	unitless	1.00 for Cabernet Sauvignon / 0.44 for Centennial Seedless	20–21

[a] 1. Rossi et al., 2009; 2. Villalta et al., 2001; 3. Rossi et al., 2008; 4. Buonassisi et al., 2017; 5. Carisse, 2016; 6. Tran Manh Sung, 1990; 7. Trilles et al., 2019; 8. Rossi and Caffi, 2012; 9. Blaeser, 1978; 10. Gessler, et al., 2011; 11. Yi et al., 2019;12. Wilcox et al., 2015; 13. Kennelly et al., 2007; 14. Boso et al., 2011; 15. Williams et al., 2007; 16. Bregaglio and Donatelli, 2015; 17. Wang et al., 2021; 18. Nogueira et al., 2018; 19. Schultz, 1992; 20. Liu et al., 2017; 21. Tang et al., 2018; 22. Gilles, 2004.

2.6. Incidence trend analysis under weather variability

Data representing the weather conditions in Beijing, Shenyang and Yantai from 2000 to 2020 were downloaded using the GLDAS Noah Land Surface Model from the NASA data portal at a spatial resolution of 0.25 × 0.25°. We used kriging interpolation to improve the data grid resolution to 0.02x0.02°. The average annual temperature and annual rainfall are the main factors that drive the occurrence and development of grape downy mildew (Caffi et al., 2016). Here, we address the change trends of these two factors and analyze the incidence distribution of grape downy mildew in multiple years (2000, 2010 and 2020).

3. Results

3.1. Sobol's analysis results

The Sobol total order index values obtained by the parameters of the SEE_DM model are shown in Fig. 2. The fluctuation of the parameter ranking gradually reduced until the number of samples was set to 40 (Fig. 2A). The hourly maximum incidence increase maintained a stable order within the ranking, suggesting its primary effect on the SEE_DM model regardless of the number of samples considered. The total order index values obtained for the grapevine varietal susceptibility indicated that the impact of this parameter on the SEE_DM model was secondary. As the number of samples changes, the sensitivities of the other parameters on model outputs exhibited weak fluctuations, and also had the same values of total order index (Table S2), except for the rain required

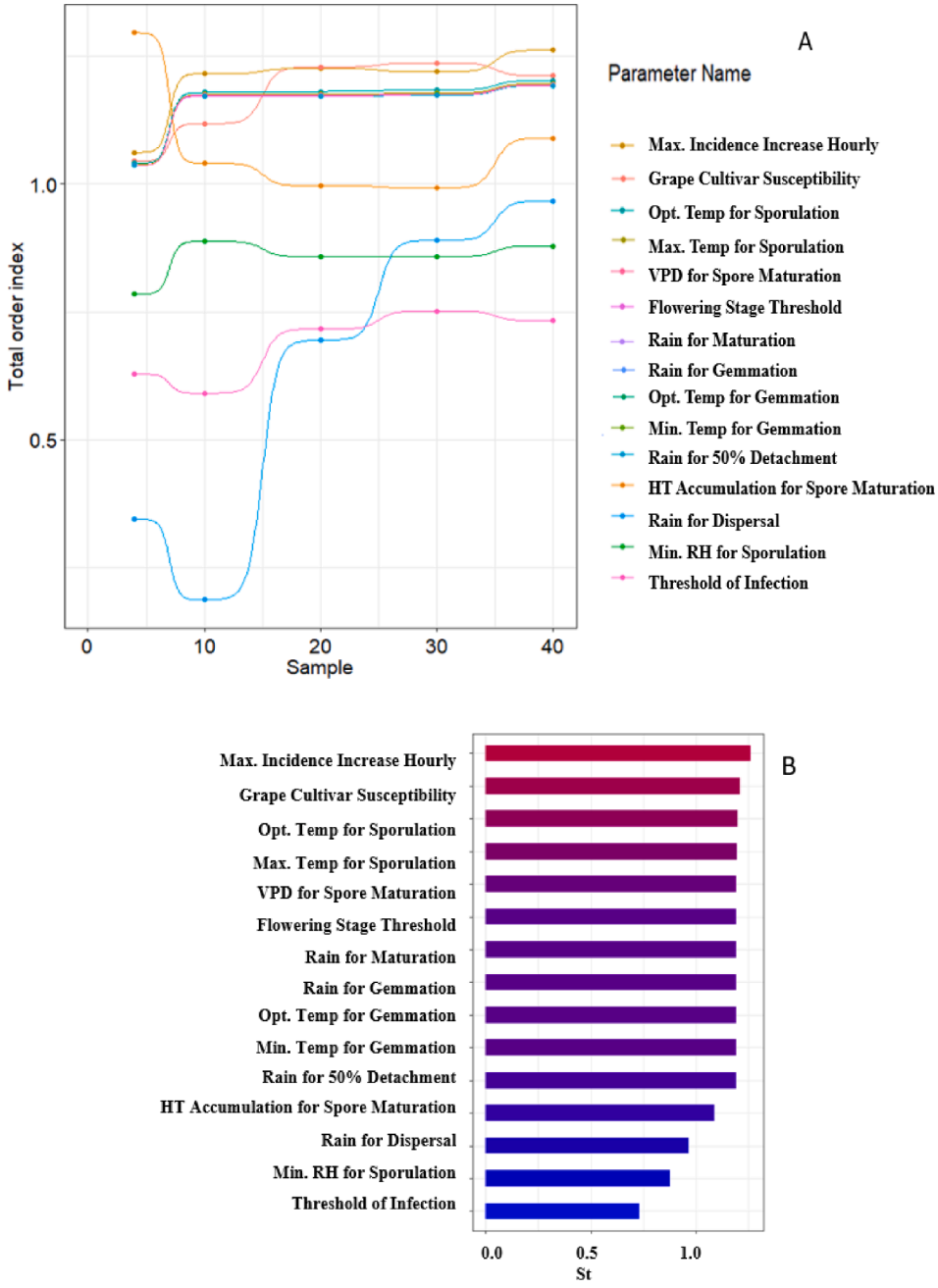


Fig. 2. Evolution of the total order sensitivity index of the parameters increased with the sample numbers (A) and when parameters were considered to have a fixed order (B, sample number = 40).

Table 3
Summary of the SEE_DM model performance when simulating grape downy mildew onset dates.

Site	Varietal	Year	Observation	Simulation	Error
Yantai	Cabernet S. [a]	2009	05–26	05–25	−1
Shenyang	Centennial S. [b]	2012	06–29	06–18	−11
		2013	07–17	07–06	−11
		2014	07–07	07–01	−6
		2015	07–01	07–14	13
		2016	06–25	06–22	−3
		2017	07–05	07–10	5
		2018	06–28	06–19	−9
		2019	06–25	06–22	−3
Beijing	Cabernet S. [a]	2011	07–25	07–18	−7
		2013	07–03	07–01	−2
		2015	07–01	06–30	−1

[a] Cabernet S.: Cabernet Sauvignon; [b] Centennial S.: Centennial Seedless.

for spores' dispersal. Moreover, the flowering phase threshold was among the parameters that reflected the impact of plant phenology on the incidence of grape downy mildew. In this case, the simulated flowering periods were similar in the three sites, leading to low parameter relevance (Fig. 2B). The maximum incidence increase hourly, grape cultivar susceptibility level and flowering phase threshold parameters were identified as key parameters to be subjected to calibration.

3.2. Model performance when simulating downy mildew onset and incidence

The dates of the first seasonal onset of downy mildew symptoms at the three sites ranged between May 26th and July 25th (Table 3). In Yantai, the date of the first appearance of disease symptoms (May 26th) was advanced by almost a month than the average dates in Shenyang and Beijing. In Shenyang, downy mildew symptoms started to be visible from the end of June until mid-July. Preventive fungicide treatments are used in Beijing to allow controlling downy mildew (Supplement Table S1), leading to setting an average date of the first disease onset in mid-July in this region, a week later than in Shenyang. Negative values

Table 4
Summary of the SEE_DM model performance when reproducing grape downy mildew incidence in the calibration and evaluation datasets.

Site	Varietal	Year	MBE	RMSE	R^2	EF
Yantai	Cabernet Sauvignon	2009	6.82	9.25	0.97	0.93
Shenyang	Centennial Seedless	2012–2019	2.29	14.14	0.91	0.71
Beijing	Cabernet Sauvignon	2011,2013,2015	6.47	15.84	0.78	0.52

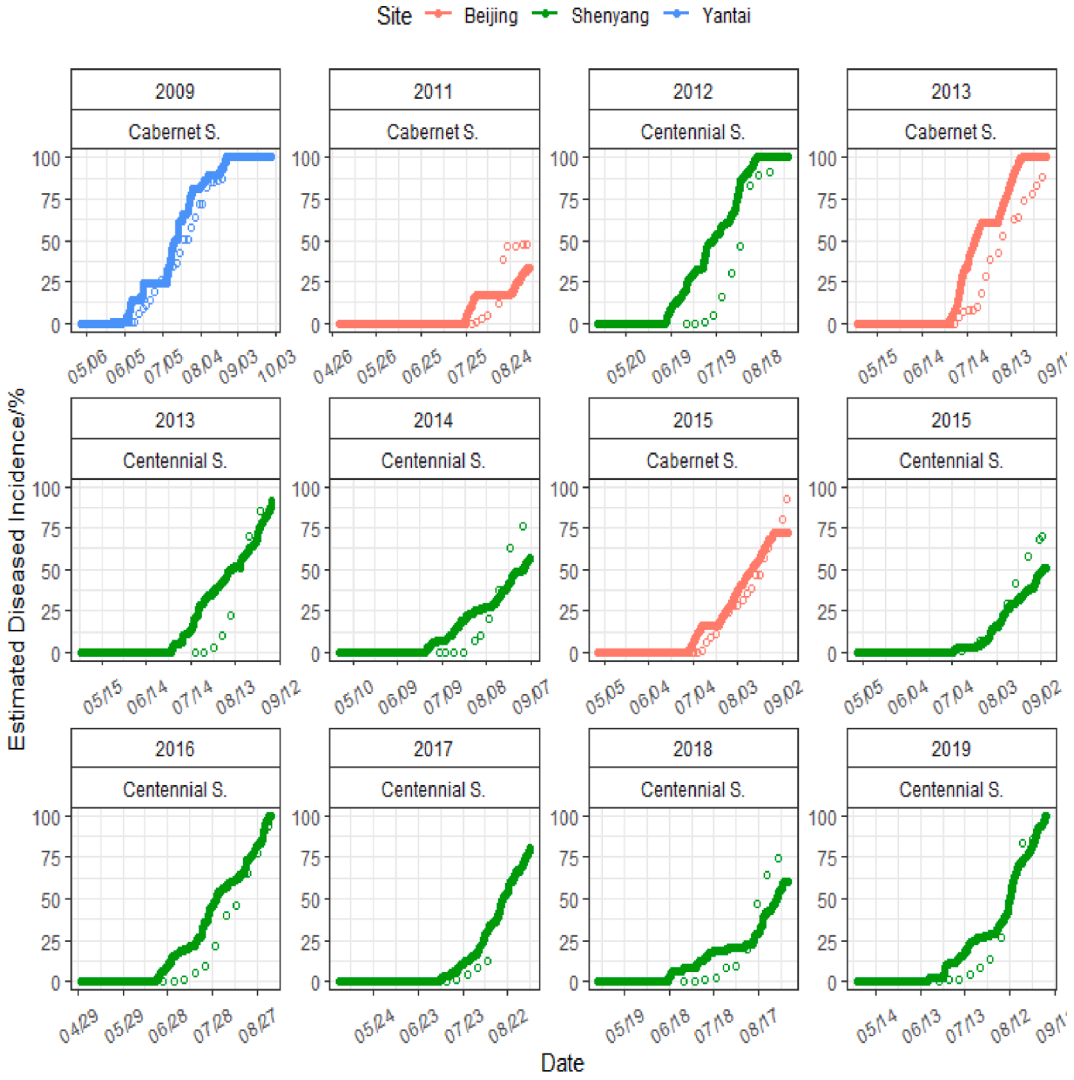


Fig. 3. Comparison of simulated (solid lines) and observed downy mildew incidence (hollow circles) on grape in experimental trials lasting several years. The charts along the horizontal axis refer to the year of analysis/the two grape varieties: Cabernet Sauvignon and Centennial Seedless.

indicate an anticipation of the onset date in simulations and viceversa. The model error ranged from −11 to 13 days. In two years (2015 and 2017) in Shenyang, the model postponed the observed symptoms onset date, whereas in 83% of cases the model predicted the disease onset with less than a week of anticipation.

Fig. 3 shows the results of the simulated and observed incidence of downy mildew in calibration and validation. The SEE_DM model simulated the dynamic of disease incidence with RMSE in the range of 2.3% − 6.8% and MBE ranging between 9.2% and 15.8% in the three analyzed sites.

The average metrics used to quantify the performance of the SEE_DM model are presented in Table 4. The R^2 values were always higher than

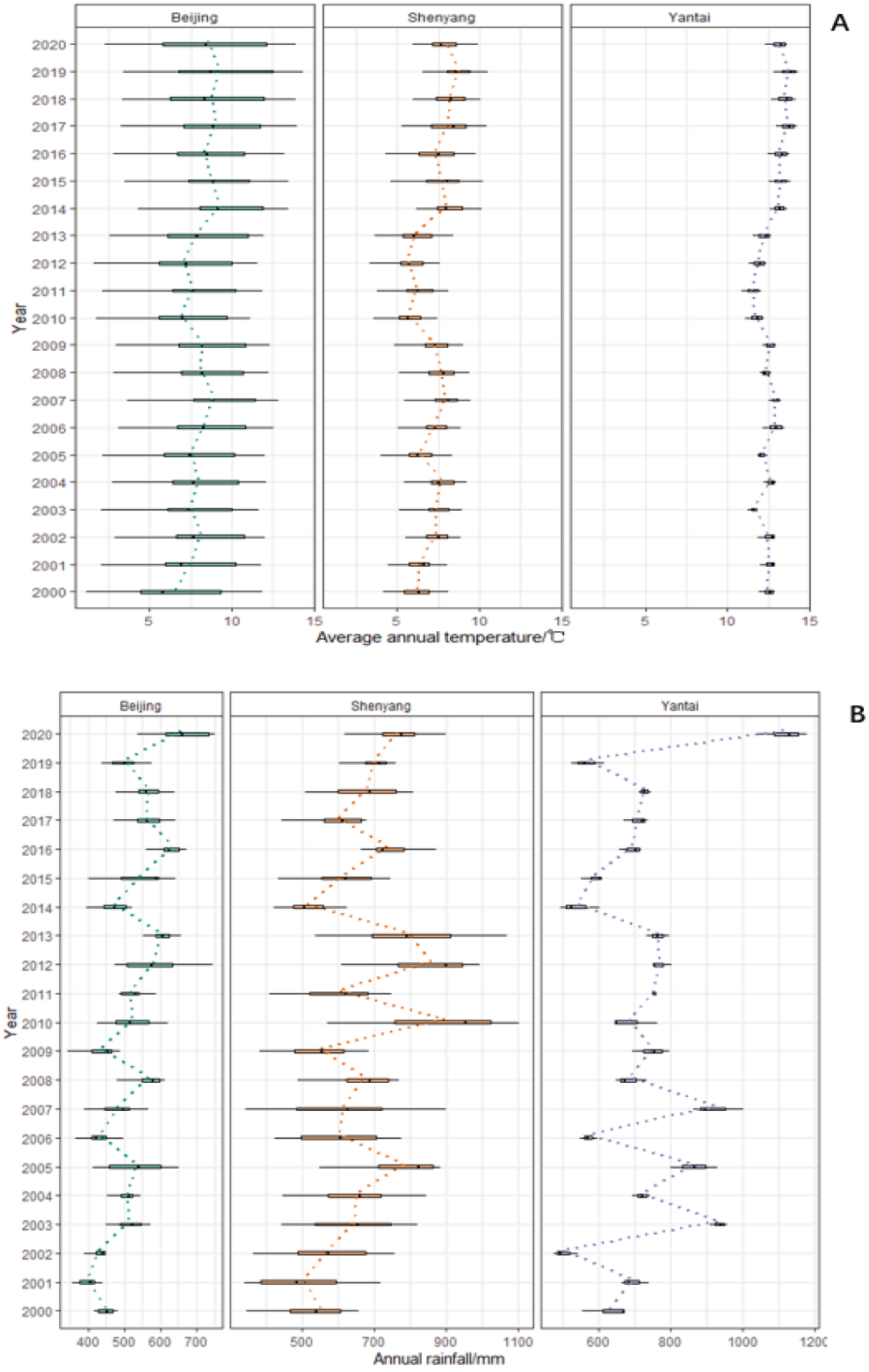


Fig. 4. Changes in the average annual temperature and annual rainfall from 2000 to 2020 at the three sites and the responses of these factors to grape downy mildew incidence.

0.78, and the EF values were all above 0.50. The best results were obtained for the Cabernet Sauvignon grapes in 2009 and 2015, with EF values above 0.90 and RMSE values lower than 10%; in contrast, the model overestimated and underestimated the incidence of downy mildew in 2013 and 2011, respectively (see supplement Table S3). On Centennial Seedless grape plants, EF values above 0.53 and R^2 values up to 0.75 were derived from 2012 to 2019. The model underestimated the incidence of downy mildew on grapes in 2013, 2014, 2015 and 2018.

3.3. Incidence trend analysis under weather variability

Two factors associated with grape downy mildew incidence under weather variability were analyzed: the average annual air temperature (Fig. 4A) and the annual rainfall (Fig. 4B). From 2000 to 2020, the average annual temperatures ranged from 6.5 to 9.3℃ in Beijing, from 5.7 to 8.7℃ in Shenyang and from 11.6 to 13.7℃ in Yantai, which resulted as the warmer region. Concerning rainfall, Shenyang and Yantai were characterized by cumulative amounts from 500 to 1000 mm/year and from 600 to 1200 mm/year, respectively. The annual rainfall in Yantai changed from 567 to 1116 mm in the last two years of the study period.

The trend analysis results obtained for the grape downy mildew incidence in three years in the study period (2000, 2010 and 2020) are shown in Fig. 5. Yantai emerged as the most suitable region for the occurrence of grape downy mildew due to its higher average annual temperatures and annual rainfall. Part of the Beijing region experienced severe grape downy mildew incidence in 2010, but most areas became suitable places for grape downy mildew in 2020 compared to the

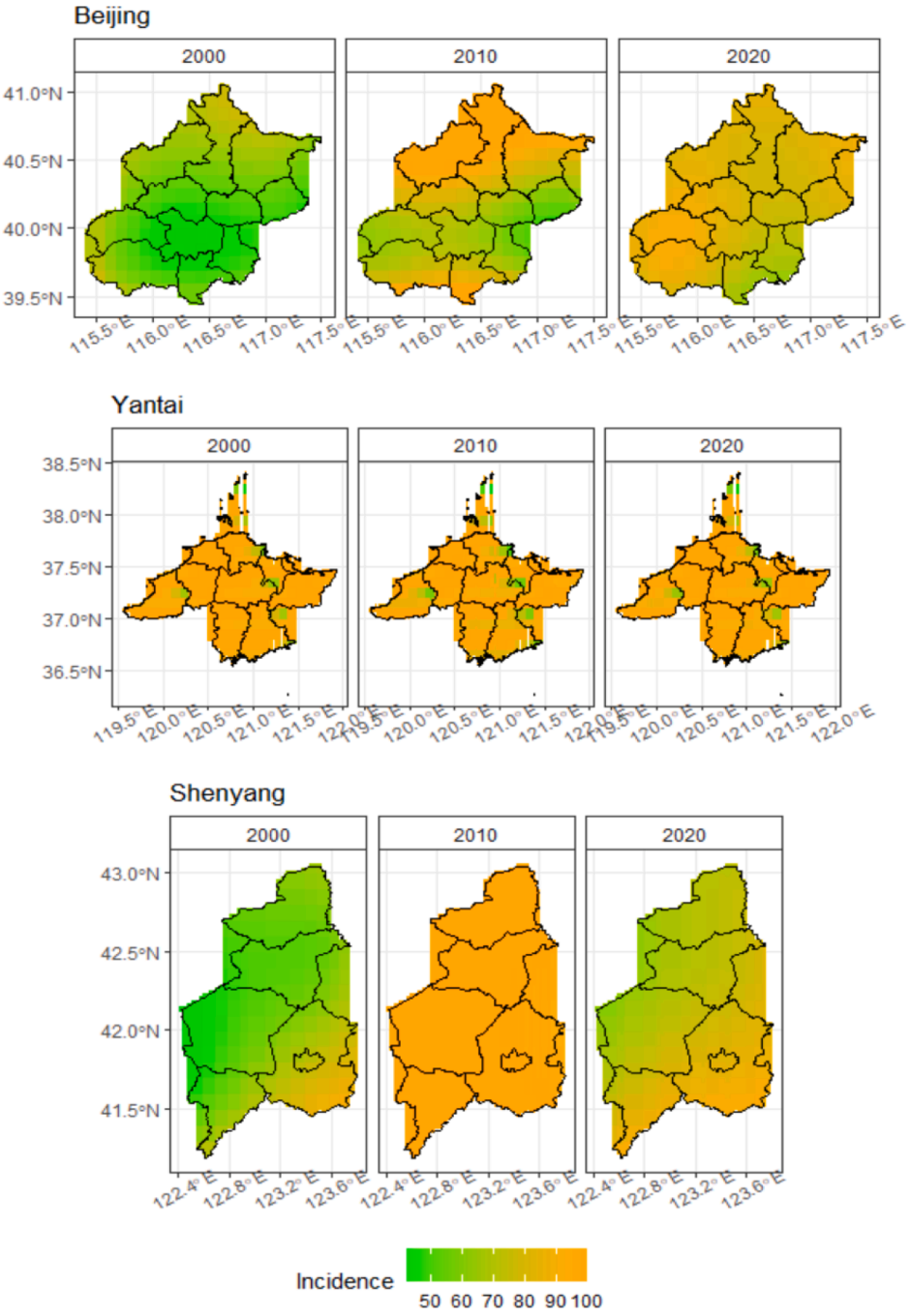


Fig. 5. Grape downy mildew incidence simulated in past years (2000, 2010 and 2020) at three sites (Yantai, Beijing and Shenyang).

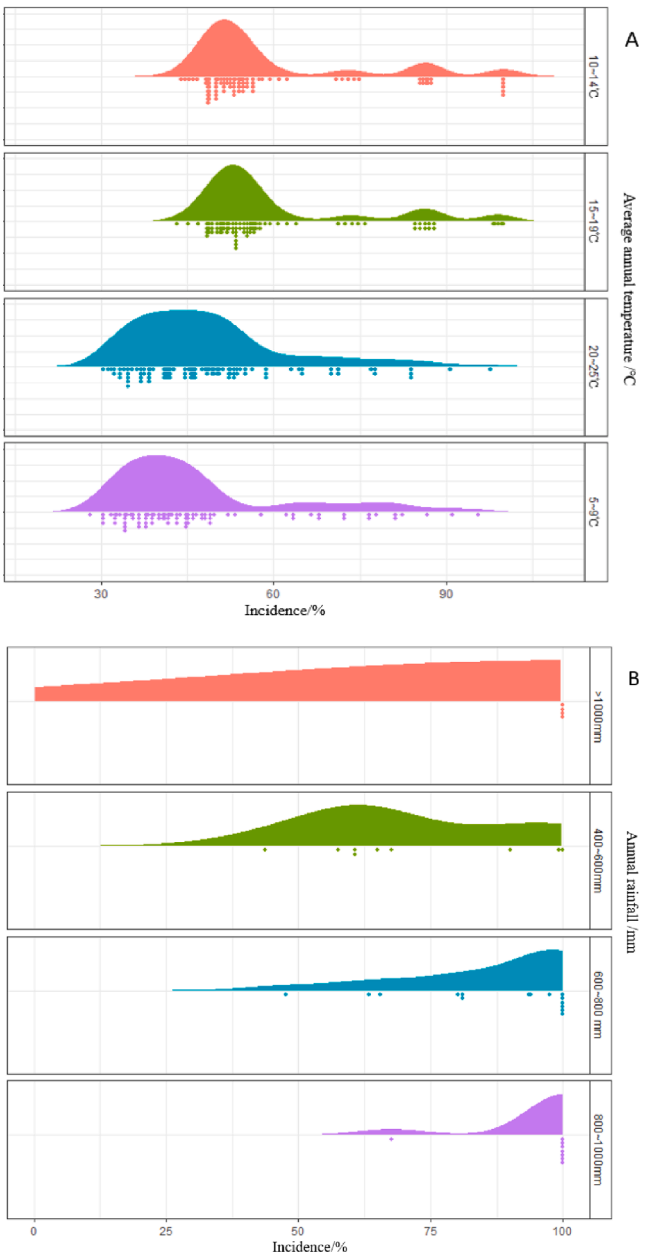


Fig. 6. Distribution of grape downy mildew incidence with regards to climate factors and weather variability.

incidence distribution in 2000. A similar incidence trend was observed in Shenyang. However, the 2010 incidence trends differed between these two sites; Shenyang had a high total annual rainfall (897.21 mm/year) and heavy grape downy mildew incidence in all areas in this year.

The relations between the distribution of grape downy mildew incidence distribution and the average temperature and rainfall changes are shown in Fig. 6. When the average annual temperature was within the 10–19 °C range, it largely contributed to the increase of grape downy mildew incidence, with respect to lower (5–9 °C range) or higher (20–25 °C range) thermal conditions (Fig. 6A). Based on this analysis, Beijing and Shenyang are becoming more suitable areas for severe grape downy mildew epidemics as the average annual temperature increases (Fig. 4A).

High rainfall is also a risk factor causing relatively heavy grape downy mildew in specific growing season. Here, rainfall over 800 mm/year was found to cause a 100% incidence rate, as shown in Fig. 6B, leading to more severe grape downy mildew incidence than either the 400 ~ 600-mm or 600 ~ 800-mm levels. The mildew incidence is thus positively related to the total annual rainfall.

4. Discussion

4.1. Parameter optimization

Reaching the grape growth stage is one of the conditions that can trigger primary downy mildew infection. Park et al. (1997) defined the grape growth stage as the time at which 5–6 leaves unfold and stated that this stage must be reached before infection can occur. However, the flowering phase is the first time at which grape downy mildew symptoms have been observed in the growing season due to the suitable temperature and sufficient rainfall conditions which occur from the end of May to mid-July in China, according to observations obtained in multisite and multi-varietal experiments. The maximum incidence increase hourly was defined in this study to describe the dynamic process by which the incidence of grape downy mildew increases. This variable has also emerged as a key parameter in other disease models, such as the rice blast model (Bregaglio and Donatelli, 2015). Varietal-specific parameter sets derived for plant and disease models are beneficial for modern agricultural technology applications at different spatial and temporal resolutions. In particular, the varietal susceptibility level to downy mildew determines the incidence of grape downy mildew in the field in the absence of controlling factors. Hence, these three key parameters were set to different values to reproduce in the multisite experimental trial.

In addition, we supposed that the biological pathogen characteristics were consistent among the three sites. The relative parameters were set to the same values in the multisite experiment, including the minimum, optimum and maximum temperatures for oospore maturation, sporangia germination and sporulation periods. We reviewed relevant papers regarding the description of the process by which the disease develops from producing mature oospores to completing the infection process. Available findings indicate similar favorable climate conditions for the production of sporangia from mature oospores (Tran Manh Sung, C. et al. 1990, Rouzet and Jacquin, 2003; Park, et al. 1997); authors agree that oospore germination occurs above a given temperature threshold (10 °C or 11 °C) in the presence of free water. Furthermore, the rainfall and degree-day accumulation at a base temperature (10 °C) were considered to be the main factors affecting oospore maturation and germination. Oospores were reported to require 170 accumulated degree-days from January 1 to reach maturation in Stuttgart, Germany (Gehmann, et al. 1987) and require an average of 92°-days in northern China based on the results derived in this study. Rosa et al. (1993) defined optimal oospore germination conditions as a consistent period above a base temperature of approximately 10 °C and sufficient rainfall of approximately 10 mm. We obtained a lower rainfall (2.5 mm) requirement for oospore germination in this study. The plant patterns and grape growth management measures are both possible reasons for these derived differences.

4.2. Model evaluation

The SEE_DM model outputs represented early warnings of the grape downy mildew onset dates and accurately reproduced mildew incidences in 12 experimental trials (in 2009 in Yantai, in 2011, 2013 and 2015 in Beijing and from 2012 to 2019 in Shenyang). Differences were obtained between the disease incidence estimations and observations when less rainfall fell during the maturation period in several years of study. However, some factors played important roles in the disease development process but were ignored in the simulations. One such factor is agronomic management, such as the pruning and removal of infected organs; this measure is usually applied to reduce disease dispersal and delay increasing disease incidence. Another such factor is the pathogen amount, which can lead to serious disease, especially as

the number of infected leaves increases.

In cases of underestimation, the main reason the model underestimated the incidence was that the optimized parameters were set fixed values in the automated calibration process to obtain the pathogen dispersal threshold caused by the amount of rainfall falling in one hour. If the threshold (set to 1.9 mm in this study, see Table 2) was over the rainfall amount that fell in one hour, the simulation assumed the environment to be unsuitable for pathogen dispersal in that hour and thus did not output a high disease increase in that short time frame. When less rainfall fell in August, the SEE_DM model generally underestimated the field data. This occurred, e.g., from 2013 to 2015 (Shenyang), when the rainfall that fell during mid- to late August was relatively low despite the high disease incidence (Supplement Fig. S1). Overall, the disease incidence estimations and observations had a good relationship ($R^2 > 0.91$ and $EF > 0.5$). Possible explanations of model underestimation of observed incidence in few cases are the use of fixed values of pathogen spores dispersal due to the amount of rainfall in one hour. If the threshold (set to 1.9 mm in this study, see Table 2) was above the rainfall amount, the model assumes non suitable conditions for pathogen dispersal, and do not increase disease incidence. When less rainfall fell in August, the SEE_DM model generally underestimated the field data. This occurred, e.g., from 2013 to 2015 (Shenyang), when the rainfall that fell during mid- to late August was relatively low, despite a high observed incidence (Supplement Fig. S1). Another possible source of error is the lack of including the effects of alternative fungicides in controlling downy mildew epidemics, e.g., those used in 2011 and 2013 in Beijing. Although two more spray applications were performed in 2011 than in 2013, under- and overestimations were observed in the simulation outputs in 2011 and 2013, respectively.

4.3. Disease trends under weather variability

The infection and development of grape downy mildew epidemics occur under suitable temperature and sufficient rainfall conditions. Due to weather change, the infection risk and development rate are being altered (Katsoulas, et al., 2021). In this study, Shenyang and Beijing were found to be turning into suitable locations for grape downy mildew infection and development due to the increase of average annual temperatures and rainfall patterns. This concern is also relevant in light of future climate scenarios. Salinari et al. (2007) suggested that a potential general advance of first disease outbreaks will occur in China from 2030 to 2080. The occurrence of other plant diseases has also largely increased in China since 1976 (Wang et al., 2021). In contrast to that, some researchers presented a decline result for some diseases under future climate change scenarios (Savary et al., 2012; Kim et al., 2015). Increased air temperatures above the optimal ranges for the pathogen and reduced rainfall during the growth season are the main reasons to explain the decline result in Brazil and different Western European countries (Juroszek and von Tiedemann, 2015). It is worth noting that the capability of pathogen infection is changed under future climate scenarios caused by temperature rise and rainfall decrease, few researches refer to this aspect.

We considered only weather variability as a determinant of incidence trend changes in this work. However, warmer temperatures not only pose as a warning for an increased plant pathogen infection risk but also signify a large change in the pathogen races present in China (Chaloner et al., 2021). In addition, we did not include potential future changes in the pathogenicity of grape downy mildew, as modeling pathogenicity using mathematical models is still difficult. However, the susceptibility levels of various varietals to grape downy mildew were considered in this work to reflect the interactions between grape plants and pathogens.

5. Conclusions

This study developed the SEE_DW model, which couples an optimized primary infection model and a new secondary infection model to warn the onset date and assess the level of damage caused by the disease. The model proved capable to provide an early warning of symptoms onset and to estimate incidence trends with reliability in three Chinese regions, with few delayed warning and incidence underestimation. The weather scenario analysis highlighted Yantai region as the most suitable area for grapevine downy mildew, and Beijing and Shenyang as becoming more conducive for the pathogen along with increased thermal and moisture patterns. Weather variability largely increases the threats posed by grape downy mildew to grape health, as was reported herein based on our analyses.

CRediT authorship contribution statement

Wang Hui: Conceptualization, Methodology. **Zhang Wei:** Methodology. **Simone Bregaglio:** Formal analysis, Supervision. **Yu Shuyi:** Methodology. **Li Xinghong:** Formal analysis, Funding acquisition. **Li Yonghua:** Methodology. **Liang Chunhao:** Methodology. **Yan Jiye:** Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compag.2022.107154.

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